

When Paraphrases Break Binding: A Lexical Cliff in Small Heads over Frozen CLIP Text Embeddings

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Abstract

Small learned heads over frozen CLIP text embeddings are routinely evaluated on held-out examples whose lexical cue variants still come from the training pool, leaving lexical generalization untested. We study this gap on a controlled compositional preference task and identify a *lexical cliff*: a per-attribute cross-attention head over frozen ViT-B/32 features loses accuracy on held-out cue variants even when they sit inside a narrow CLIP-text neighbourhood (cosine $[0.87, 0.97]$ to training) and the visual input is unchanged. A substitution probe that swaps only the text embedding lifts the worst-failing token from 0.500 to 0.893, and a 2×2 encoder factorial shows that cliff closure tracks the text encoder rather than the visual front-end. The cliff has two distinguishable mechanisms—a cosine-correlated decline on motion verbs ($r=0.627$, $n=12$) and a class-bimodal pattern on size and speed—and replicates across OpenCLIP-LAION-2B and SigLIP-WebLI with encoder-family-dependent token identity. Scaling to ViT-L/14 closes most but not all of the cliff, indicating the threshold is set by text-encoder geometry rather than raw cosine. We release CF-PrefBench v4 and recommend lexical-held-out splits with cosine stratification for evaluating small learned heads over frozen CLIP text.

1 Introduction

Frozen pretrained text encoders—CLIP (Radford et al., 2021) chief among them—have become the default substrate for compositional vision-language systems: a small learned head consumes text embeddings and grounds them in visual features. Held-out evaluation in this regime typically randomises over examples but reuses the training-time paraphrase vocabulary, so the lexical surface of test instructions stays close to training. Prior compositional benchmarks (Thrush et al., 2022; Ray et al., 2023; Hsieh et al., 2023) focus on hard-negative

captions far from the training distribution, but the question of what happens to held-out cue variants that sit *inside* a narrow lexical neighbourhood—tokens whose CLIP-text cosine to training is in $[0.87, 0.97]$, the zone a practitioner would intuitively consider safe—is largely unasked. We pose it on a controlled compositional preference task and find a sharp answer.

We construct CF-PrefBench v4 with held-out lexical cue variants disjoint from training and spanning CLIP-text cosine $[0.87, 0.97]$, train a per-attribute cross-attention head over frozen CLIP patches (LIB v0), and ask: does it generalize inside this “close-enough” neighbourhood, and where does failure localise? Accuracy collapses sharply. A substitution probe (text embedding swapped, video fixed) recovers the worst-failing token from 0.500 to 0.893, and a 2×2 factorial swapping visual and text encoders between B/32 and L/14 confirms only the text-side swap closes the cliff. The failure has two distinguishable mechanisms—a cosine-correlated decline on motion verbs and class-bimodal defaulting on size and speed—and replicates across OpenCLIP-LAION-2B and SigLIP-WebLI with encoder-family-dependent token identity. Our goal is not a stronger preference model but to expose a lexical-generalization blind spot that held-out-example evaluation misses.

Contributions. (1) A lexical cliff in compositional binding over frozen CLIP text embeddings, causally localised to the text encoder by a 2×2 factorial within OpenAI CLIP and replicated across two additional contrastive systems (OpenCLIP-LAION-2B and SigLIP-WebLI). (2) Characterisation of two empirically distinguishable cliff mechanisms: cosine-correlated decline on motion verbs and geometry-mediated class-defaulting on size and speed, with encoder-family-dependent token identity. (3) CF-PrefBench v4 (7 binding axes,

7,749 rows, cosine-stratified held-out cue variants), released as an anonymous artifact along with auxiliary diagnostic scripts (§5) for compositional preference models with separable signal pathways.

2 Related Work

Compositional and lexical generalization. A long line of work documents that pretrained encoders cluster training-time lexical neighbourhoods and that compositional swaps cause sharp accuracy drops (Lake and Baroni, 2018; Kim and Linzen, 2020; McCoy et al., 2019; Kaushik et al., 2020). In vision-language, Winoground (Thrush et al., 2022), CoLA (Ray et al., 2023), and SugarCrepe (Hsieh et al., 2023) report near-chance accuracy on compositional swaps and construct hard-negative captions with matched text statistics. These benchmarks focus on captions *far* from the training surface; our setting differs in two respects. First, we evaluate cue variants that sit *inside* a narrow lexical neighbourhood (CLIP-text cosine $[0.87, 0.97]$ to training) and report accuracy as a continuous function of cosine, exposing a small-encoder cliff that binary swap metrics hide. Second, our substitution probe (text embedding swapped, video fixed) gives a controlled causal localisation of failure to the text side—an intervention not standard in compositional benchmarks.

Probing frozen text encoders. CLIP-text-side analyses show the encoder’s geometry is sensitive to surface form in ways that affect downstream behaviour: bag-of-words effects (Yuksekgonul et al., 2023), order insensitivity (Thrush et al., 2022), limited compositional structure (Lewis et al., 2024), and a representational bottleneck that loses information about attributes, relations, and counts when single captions are pooled into a single vector (Kamath et al., 2023). Recent work removes the vision tower entirely and runs retrieval as text-to-text matching to bypass this bottleneck (Ntinou et al., 2025); within the standard CLIP pipeline, low attribution to relations and attributes has been identified (Li et al., 2024) and individual token weights have been re-scaled to give finer control over text embeddings (Kim et al., 2024). We add a preference-task perspective: the same family of limitations manifests as a backbone-specific cosine threshold for downstream binding, and the threshold moves with text-encoder identity (per the 2×2 factorial) rather than with raw cosine. We additionally introduce a small sanity-diagnostic battery for

multi-pathway preference models (§5), extending shortcut-probing precedents in NLP (McCoy et al., 2019; Kaushik et al., 2020; Gardner et al., 2020) to vision-language preference models.

Preference learning and language-conditioned

RL. Preference learning from paired trajectories (Christiano et al., 2017; Lee et al., 2021; Hejna et al., 2024) and language-conditioned robotics resources (Sermanet et al., 2024; Padalkar et al., 2024) do not stratify evaluation by lexical cosine to training; the cliff we report would be invisible under their protocols. We adopt the pairwise BCE recipe common to this line but evaluate on a lexical-held-out split with cosine-stratified analysis.

3 CF-PrefBench v4 and the LIB v0 Method

Task. Each example is a triple (v_A, v_B, t) of two video trajectories differing on one binding axis (color, object, action, spatial, size, motion_sequence, or speed) and a natural-language instruction t . The task is to predict $y \in \{A, B\}$; we measure per-row accuracy and flip-group preference accuracy (PFA).

Benchmark. CF-PrefBench v4 extends the four-axis v3 release with three new axes (size, motion_sequence, speed) using a deterministic rendered preference simulator producing 24-frame trajectories at 192×144 resolution. It comprises 7,749 video-pair examples across the seven axes plus an `impossible_premise` diagnostic split (sizes in Appendix E, Table 9). The rendering is intentionally minimal: by removing photorealism and clutter, we eliminate visual recognition as a confound, so any failure must localise to the binding mechanism itself.

The critical evaluation split is the held-out-lexical split (`test_heldout_lexical`): instructions use cue variants that never appear in training. Train and held-out pools are disjoint by construction and chosen so that held-out cue variants span the CLIP-text cosine zone $[0.87, 0.97]$ to training-pool instructions on ViT-B/32 (per-axis pools in Appendix E, Table 10). On motion_sequence we add a fourth held-out verb, *scoot*, to probe the cliff transition zone. For the cosine-monotonic analysis (§4.1) we additionally evaluate a stratified pool of 12 motion verbs that spans the cliff cosine zone.

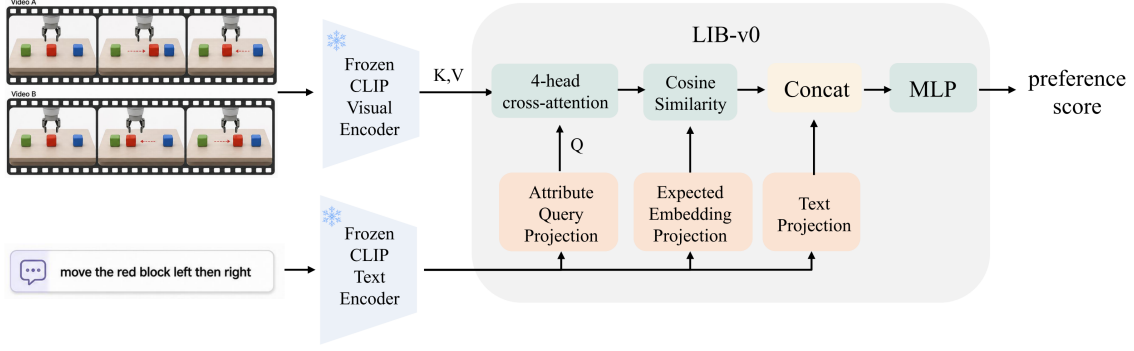


Figure 1: LIB v0 architecture. Frozen CLIP encoders (snowflake) feed visual patches (K, V) and three independent text projections. The attribute query projection produces queries (Q) for the 4-head cross-attention; cosine similarity against the expected embedding projection gives a 4-d binding score per video; a 16-d text projection plus both per-video scores are concatenated and passed through an MLP. Cross-attention and cosine similarity use shared weights across videos A and B. A reconstruction auxiliary (Appendix H) prevents query collapse.

LIB v0. LIB v0 is a per-attribute cross-attention head over frozen CLIP visual patches (Figure 1; full architecture in Appendix H). Four attribute queries projected from the CLIP text embedding attend over visual patch tokens; cosine similarity against a per-attribute “expected” projection gives a 4-d binding score concatenated with a 16-d text projection and passed through a 64-hidden-unit MLP. A reconstruction auxiliary prevents query collapse. Trainable parameters total $\sim 0.6\text{M}$ (B/32) / $\sim 0.9\text{M}$ (L/14); training uses AdamW (60 epochs, 3 seeds) with preference BCE, reconstruction CE, counterfactual margin, and paraphrase stability terms.

Baselines and evaluation. We compare: (i) an engineered centroid baseline (173-dim hand-engineered features through a 2-layer MLP; 0.929 PFA on v3 `test_heldout_color`); (ii) LIB v0 on B/32 and L/14; (iii) the Phase 3 hybrid (case study in §5). Zero-shot GPT-4o (OpenAI, 2024) and Qwen2.5-VL-7B (Bai et al., 2025) decodability results are reported in Appendix L. Learned methods use three seeds. §4 stratifies by CLIP-text cosine to mean training instruction. Load-bearing claims rest on effect sizes ($\Delta \geq +0.226$) exceeding seed-level variance. Cliff tables use either 3-seed or 8-seed protocols depending on sample-size considerations; each table caption specifies the seed count and token set used.

4 The Lexical Cliff

Scope: cliff appears on cue-replacement axes only. The lexical cliff is not uniform across CF-PrefBench’s seven binding axes. On v3 (4 axes)

and v4 (3 axes), the cliff signature—a sharp held-out-lexical drop with strict PFA=0 on at least one held-out token—appears on *action* (v3), *motion*, *size*, and *speed* (v4), and is absent on *color*, *spatial*, and weak on *object* (v3). The partition is predicted by a single structural distinction: whether the held-out cue variant replaces the binding cue itself or only the wrapper text around an unchanged cue.¹ On the v3 action axis, the cliff signature is structurally identical to v4 size SMALL / speed FAST: held-out verbs in one direction class collapse to 0.10–0.19 row accuracy while the opposite class lifts to 0.81–0.86, with PFA = 0.000 across 3 seeds (Appendix G). The per-token cliff analysis below uses the v4 axes because v4 added cosine-stratified held-out tokens needed to chart the cliff at the token level.

4.1 Motion Verbs: A Cosine-Correlated Decline

We test LIB v0 on `motion_sequence_test_heldout_lexical`, which substitutes verbs {shift, convey, transit, scoot} for training verbs {move, drag, push}. Held-out and training paraphrases share identical direction tokens (“left then right” or “right then left”); only the verb changes. Table 1 shows per-row accuracy alongside the CLIP-B/32 cosine of each held-out instruction to the mean training instruction. Accuracy is correlated with cosine, with the strongest

¹On color, held-out instructions swap “press the cyan button” for “engage the cyan switch”—the cue token *cyan* is preserved. On action, “open the drawer” becomes “extract the drawer”—the verb *is* the cue and is fully replaced.

Verb	cos(B/32)	Accuracy (3-seed)
shift	0.968	0.917 $\pm$ 0.017
convey	0.938	0.929 $\pm$ 0.000
scoot	0.927	0.679 $\pm$ 0.058
transit	0.915	0.500 $\pm$ 0.000

Table 1: ViT-B/32 motion-verb cliff on the 4-verb cliff probe. Effect-size pattern: the worst verb drops 0.417 below the best, an order of magnitude above seed-level noise ($\sigma \leq 0.058$). Zero variance on *transit* reflects a fully collapsed predictor: all three seeds map the verb to the same wrong direction. The cosine-correlation relationship is quantified on an expanded pool of 12 motion verbs in Figure 4(a); per-token values in Table 7 (Appendix C). Per-seed values for *shift/convey*: see Appendix C and Appendix K, which use a matched seed subset (1/2/3) under the cross-family recipe; small numerical differences do not affect any qualitative claim.

collapse at the lower-cosine end. The 4-token probe is not intended as a strict monotonicity test; the cosine-correlation relationship is quantified on the expanded 12-verb pool below ($r=0.627$). We report row-level accuracy throughout; the strict flip-group PFA score collapses to zero when one verb (*transit*) fails systematically, as its aggregation is per-flip-group.

Cosine-correlated decline on an expanded verb pool. To quantify the cosine-correlation relationship on a larger pool than the four cliff-probe verbs, we evaluate the published LIB-B/32 checkpoints (seeds 1, 2, 3) on an expanded set of 12 motion verbs chosen to span the CLIP-B/32 cosine range [0.85, 0.99], controlled for BPE length, part-of-speech, and Zipf frequency (Appendix B). Over $n=12$ verbs, Pearson $r=0.627$ with 95% bootstrap CI [0.158, 0.916] (excludes zero); a partial Pearson controlling for $\log(\text{Zipf})$, polysemy, and BPE length is $r_{\text{partial}}=0.576$, indicating cosine remains the dominant predictor after these confounders.

The failure is text-side. A substitution probe—swapping the held-out instruction’s CLIP text feature for a training-verb instruction with identical direction tokens, video and label unchanged—recovers *transit* from 0.500 to 0.893 and *scoot* from 0.679 to 0.917 (Appendix A); recovery scales inversely with cosine. The model reads direction from the video; the projection from a far-from-training text embedding to attribute queries fails. This localises failure to the upstream pipeline (text encoder combined with the learned text-projection layer in LIB); the 2×2 factorial in §4.3 further nar-

Verb	cos(B/32)	B/32	cos(L/14)	L/14
shift	0.968	0.917	0.925	0.905
convey	0.938	0.929	0.918	0.929
scoot	0.927	0.679	0.875	0.905
transit	0.915	0.500	0.876	0.917

Table 2: Motion-verb cliff under two backbones. Both B/32-cliff verbs recover to ≥ 0.905 on L/14 despite their L/14 cosines to training being lower than under B/32.

rows the locus to the text encoder itself.

4.2 ViT-L/14 Closes the Cliff

If the cliff stems from B/32’s text-embedding geometry, a larger CLIP variant should reduce or remove it. We re-extract features with ViT-L/14 and re-train LIB v0 from scratch for three seeds under the same recipe (batch reduced to 16). Table 2 reports both backbones; the two B/32-cliff verbs gain +0.226 (*scoot*) and +0.417 (*transit*) under L/14. Crucially, this happens despite L/14 placing both verbs at lower cosine to training than B/32 did. *There is no universal cosine threshold for binding*; the threshold depends on the text encoder’s embedding geometry, and raw cosine alone is insufficient to predict downstream binding accuracy.

4.3 Causal Isolation: A 2×2 Encoder Factorial

The L/14 closure leaves an attribution question open: does cliff closure require the larger *text* encoder, the larger *visual* encoder, or both? We resolve this with a 2×2 factorial that independently swaps the two encoders between B/32 and L/14, training three seeds per cell under the same recipe; a learnable linear pre-projection maps the text-embedding dimension to a fixed 512 before LIB consumes it, controlling for text-dim differences. Table 3 reports cliff-class mean accuracy per cell.

The pattern is unambiguous (Table 3, Figure 2). Text-side swap (B-L) closes the cliff to 0.827 ± 0.029 , within seed noise of the full L/14 upgrade (0.803 ± 0.012). Visual-side swap (L-B) leaves it intact (0.611 ± 0.035), indistinguishable from baseline (0.633 ± 0.016). In this factorial, swapping the text encoder is sufficient to close the cliff, whereas swapping only the visual encoder is not. Combined with §4.1’s substitution probe, the frozen CLIP-B/32 text encoder is the primary locus of failure within this factorial; the head’s learned text-projection layer cannot fully recover from upstream

Cell	Visual	Text	Cliff acc. (3 seeds)
B-B (baseline)	B/32	B/32	0.633 ± 0.016
B-L	B/32	L/14	0.827 ± 0.029
L-B	L/14	B/32	0.611 ± 0.035
L-L (full upgrade)	L/14	L/14	0.803 ± 0.012

Table 3: Cliff accuracy under the 2×2 {visual encoder} $\times$ {text encoder} factorial, mean $\pm$ std over 3 seeds. Replacing only the text encoder (B-L) closes the cliff, whereas replacing only the visual encoder (L-B) does not, supporting text-side localisation. Each value is the equal-weight mean across the 12 cliff probe tokens; seed sets $\{1, 2, 3\}$ for B-B and L-L, $\{42, 123, 2024\}$ for B-L and L-B. The $n=3$ value is within 0.014 of the canonical $n=8$ protocol (Table 5); protocol details in Appendix C.

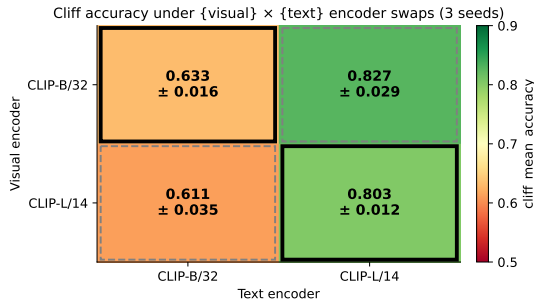


Figure 2: Cliff accuracy heatmap under the 2×2 encoder factorial. Diagonal cells (B-B, L-L) show baseline and full upgrade; off-diagonal cells (B-L, L-B) isolate the contribution of each encoder. Black solid borders mark cells where the text encoder is B/32 (cliff present); gray dashed borders mark cells where the text encoder is L/14 (cliff closed). The pattern visualises the text-side localisation: cliff closure tracks the text encoder, not the visual encoder.

geometry that places held-out tokens on the wrong side of the class-discriminating direction.

4.4 Cross-Family Validation

The 2×2 factorial isolates the text encoder *within* the OpenAI CLIP family. To test whether the mechanism generalises across contrastive families, we re-train LIB v0 with both encoders replaced by OpenCLIP ViT-B/32 trained on LAION-2B²—an architecturally matched, independently trained contrastive pipeline (cell OC-OC; 3 seeds, identical recipe).

The cliff replicates but its *token identity* swaps (Figure 3). B-B’s two strongest cliffs close under LAION-2B (size SMALL $0.444 \rightarrow 0.929$; motion

²We use the laion2b_s34b_b79k checkpoint from the open_clip library.

Class (tokens)	B-B	OC-OC	SG-SG
Motion above-cliff	0.887	0.679	0.661
Motion below-cliff	0.595	0.839	0.780
Size BIG	0.925	0.972	0.929
Size SMALL	<i>0.444</i>	0.929	0.817
Speed FAST	<i>0.115</i>	0.321	0.619
Speed SLOW	0.829	0.639	0.667

Table 4: **Cross-family validation (matched $n=3$ protocol)**. Mean cliff-class accuracy under three contrastive systems: B-B = OpenAI CLIP-B/32 (seeds 1/2/3), OC-OC = OpenCLIP-B/32-LAION-2B (seeds 42/123/2024), SG-SG = open_clip ViT-B-16-SigLIP / WebLI (seeds 42/123/2024). **All values use $n=3$ seeds and 2 tokens per class for protocol-matched cross-family comparison**; canonical B-B estimates at $n=8$ seeds $\times$ 4 tokens (Table 5) are 0.485 (SMALL) and 0.159 (FAST), tighter but not directly comparable across encoder families. *Italic*: B-B class mean below chance (≤ 0.5). **Bold**: tokens where the cliff status swaps relative to B-B (B-B ≤ 0.5 AND alternative system > 0.7 , or vice versa). Per-token tables in Appendix K.

below-cliff $0.595 \rightarrow 0.839$); speed FAST partially recovers ($0.115 \rightarrow 0.321$). Meanwhile *shift* (B-B’s easiest above-cliff verb) drops from 0.917 to 0.488 and *sluggishly* drops from 0.960 to 0.603—new partial cliffs at weaker magnitude.

A second contrastive family: SigLIP. To test whether the token-identity-swap pattern extends beyond CLIP-style softmax-contrastive encoders, we additionally train LIB v0 with both encoders replaced by open_clip ViT-B-16-SigLIP (Zhai et al., 2023) (Google WebLI weights), a contrastive system that differs from OpenAI CLIP along several confounded axes (sigmoid loss, WebLI data, patch-16 grid, squash-resize preprocessing); we treat it as a holistic family swap rather than isolating any single design choice. The LIB head architecture is otherwise identical, with a $768 \rightarrow 512$ pre-projection matching the B-L cell of §4.3.

The pattern under SigLIP mirrors the LAION-2B finding rather than abolishing the cliff: three B-B failure tokens close (*miniature* $0.397 \rightarrow 0.921$, *petite* $0.492 \rightarrow 0.714$, *briskly* $0.048 \rightarrow 0.865$; per-token in Appendix K), but *sluggishly* becomes a new cliff token ($0.960 \rightarrow 0.333$) and *speedily* remains below chance ($0.183 \rightarrow 0.373$).

Takeaway. The cliff is not tied to a single CLIP checkpoint: across three contrastive systems (Figure 3), a failing region always exists, but *which* tokens fall into it is encoder-family dependent. Combined with §4.3’s within-family localisation, this

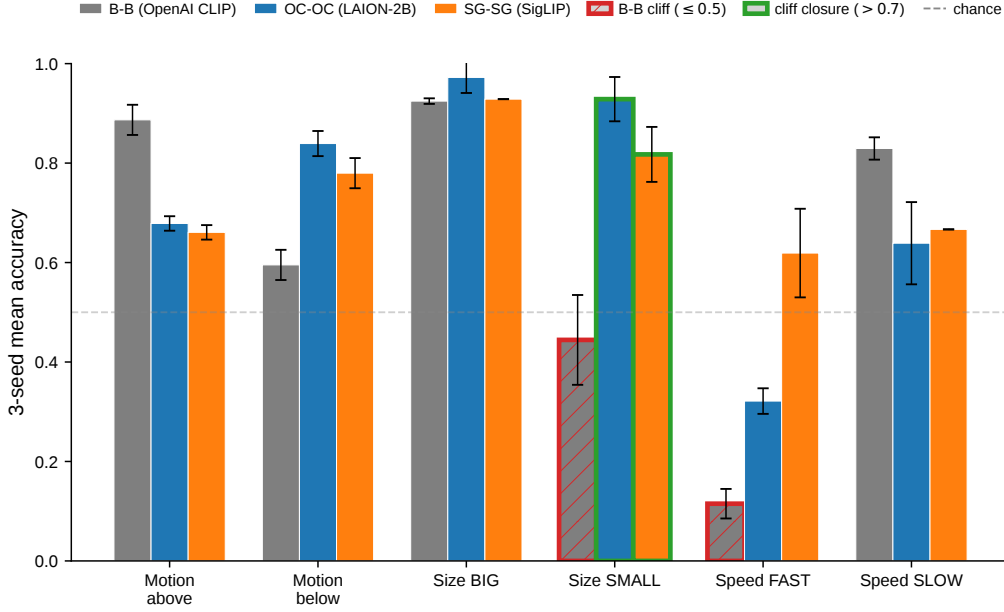


Figure 3: **Cross-family validation of the lexical cliff.** Per-class 3-seed mean held-out accuracy on the six cliff classes under three contrastive systems: OpenAI CLIP-B/32 (B-B, gray), OpenCLIP-LAION-2B (OC-OC, blue), and `open_clip` ViT-B-16-SigLIP / WebLI (SG-SG, orange). Error bars: one standard deviation across seeds (B-B: 1/2/3; OC-OC and SG-SG: 42/123/2024). Red hatched borders mark B-B cliff cells (mean ≤ 0.5); green borders mark cells where an alternative system closes a B-B cliff (B-B ≤ 0.5 and alternative > 0.7). Per-token breakdown in Appendix K.

refines the geometry-mediated defaulting account in §4.5: the head’s class-discriminating direction is fixed by training, but which held-out tokens fall on which side of that direction is mediated by upstream contrastive geometry. The cross-family swap re-routes which tokens fall into the failing region, but does not change the fact that a failing region exists.

4.5 Size and Speed: A Class-Bimodal Cliff

We evaluate the same $n=8$ -seed trained models on sixteen held-out tokens spanning size and speed (Table 5, per-token in Appendix C): {colossal, gigantic, enormous, vast} (BIG), {miniature, petite, tiny, minute} (SMALL), {briskly, speedily, swiftly, hastily} (FAST), {sluggishly, gradually, leisurely, languidly} (SLOW). Every token lies inside the B/32 cliff zone (cosine-to-training $\in [0.89, 0.97]$) and is absent from training. No retraining; the original 8-seed checkpoint pool is used zero-shot. 95% paired-by-seed bootstrap CIs are reported.

Two mechanisms emerge (Figure 4). On motion, accuracy is a smooth function of cosine. On size and speed, held-out tokens within the same class share near-identical cosines but split into two regimes: BIG and SLOW succeed (B/32 cell means

0.924 and 0.792); SMALL and FAST collapse (0.485 and 0.159). Within each class cosine cannot predict accuracy; class membership can. The single worst FAST token, *briskly*, sits at 0.048 (8-seed mean for that one token; the class mean over four FAST tokens is 0.159), well below chance—indicating systematic anti-correlation: the model picks the slower video even when told “move the block briskly.” We interpret this as *geometry-mediated class-defaulting*: the head learns a class-discriminating direction from training tokens, and held-out tokens whose CLIP embeddings land on the wrong side of this direction trigger the head to default to the opposite class. Which tokens land on which side is set by upstream text-encoder geometry, not by a fixed prior over BIG/SLOW per se; §6 unpacks the mechanism. The motion axis lacks an obvious default (direction tokens are symmetric in training), so its cliff degrades smoothly. L/14 largely closes both mechanisms but only partially on FAST (cell mean 0.159 $\rightarrow$ 0.545); the small regression on size BIG (-0.054 in Table 5) reflects ceiling saturation of an already-passing class rather than a new failure mode.

Axis	Class	B/32 ($n=8 \times 4$ tok)	L/14 ($n=8 \times 4$ tok)	Δ
Motion	above-cliff	0.902 [0.884, 0.920]	0.922 [0.908, 0.935]	+0.020
	below-cliff	0.621 [0.592, 0.650]	0.917 [0.904, 0.929]	+0.297
Size	BIG	0.924 [0.920, 0.929]	0.870 [0.840, 0.900]	-0.054
	SMALL	0.485 [0.432, 0.540]	0.802 [0.756, 0.847]	+0.317
Speed	FAST	0.159 [0.100, 0.214]	0.545 [0.487, 0.617]	+0.386
	SLOW	0.792 [0.766, 0.820]	0.839 [0.832, 0.845]	+0.046

Table 5: Three-axis $\times$ two-architecture cliff matrix on the *expanded* held-out token pool (4 tokens $\times$ 8 seeds = 32 datapoints per size/speed cell; 2 tokens $\times$ 8 = 16 for motion sub-classes). Brackets give paired-by-seed 95% bootstrap CIs (1000 resamples). **Bold** = cliff cells under B/32. L/14 fully closes the cliff on motion and largely on SMALL (0.485 $\rightarrow$ 0.802); on FAST, recovery is partial (+0.386) with per-token heterogeneity within the FAST class (e.g., *hastily* 0.20 $\rightarrow$ 0.99 but *briskly* 0.05 $\rightarrow$ 0.32; Appendix C).

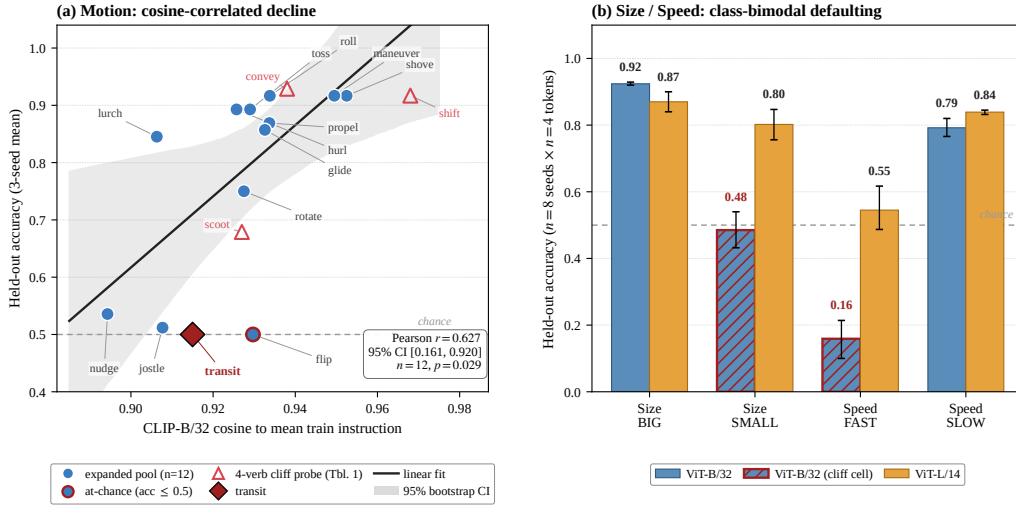


Figure 4: Two cliff mechanisms. (a) Motion verbs: cosine-correlated trend on the canonical expanded pool ($n=12$ verbs, Pearson $r=0.627$, 95% bootstrap CI [0.158, 0.916], excludes zero). The four cliff-probe verbs of Table 1 are highlighted; *transit* (red diamond) is the only B/32 below-chance cliff verb. (b) Size and speed ($n=8$ seeds $\times$ $n=4$ tokens per cell): tokens within a class cluster at near-identical cosines but accuracy splits along the head’s class-discriminating direction over held-out text (BIG/SLOW pass, SMALL/FAST fail). ViT-L/14 largely closes the SMALL cliff (0.485 $\rightarrow$ 0.802) and partially recovers FAST (0.159 $\rightarrow$ 0.545). Bars show $n=8$ seeds $\times$ $n=4$ tokens mean with 95% paired-by-seed bootstrap CIs; hatched red borders mark B/32 cliff cells. See §6.

4.6 Robustness Across Alternative Heads

The cliff is not specific to LIB v0’s architecture. We re-train three alternative heads—a bilinear matching head $t^\top Wv$, a 2-layer MLP over $[t \parallel \bar{v}]$, and a cross-attention head without attribute queries (3 seeds each; recipe matched to LIB v0)—over the same frozen B/32 features. All three reproduce the above-vs-below gap within ± 0.05 effect-size; the bilinear gap reaches significance ($p=0.004$), MLP ($p=0.058$) and X-attn ($p=0.079$) are marginal under $n=3$. We treat effect-size consistency as the load-bearing claim and flag the $n=3$ significance caveat as a limitation; full table, figure, and per-head training details are in Appendix D.

4.7 Generalization Probes

We additionally test two generalization regimes (full details in appendices). **Decodability by larger VLMs:** zero-shot GPT-4o on the same cliff test set resolves the failed tokens (≥ 0.81 on size SMALL, 0.68 on speed FAST; +0.34 to +0.60 over LIB-B/32), while Qwen2.5-VL-7B sits near chance, suggesting the cliff tokens are decodable by at least one stronger zero-shot VLM and that the failure is specific to small learned heads over frozen CLIP-text features (Appendix L). **Out-of-domain rendered video:** on ManiSkill PushCube (Franka cube-pushing, zero-shot, no retraining), the class-bimodal speed cliff partially re-emerges on B/32: held-out FAST adverbs collapse to 0.33 while SLOW adverbs hold at 0.62, and a text-side sub-

stitution probe recovers FAST to 0.64 ($\Delta=+0.31$), matching the simulator. ViT-L/14 does not transfer to this domain, so its cliff is untestable (Appendix M). We frame this as a bounded generalization signal rather than a real-world transfer claim.

5 Sanity Diagnostic Battery

Alongside the main investigation we developed a small set of sanity probes for compositional preference models with separable signal pathways, included as a methodological aid rather than an independent contribution. During development, these probes caught an earlier “positive” result as a centroid pass-through: a hybrid LIB+centroid model scored 0.905 ± 0.034 on v3 `test_heldout_color`, but six probes showed the LIB path contributed nothing—SC-4 (zero centroid) collapsed accuracy from 0.929 to 0.167, while SC-5 (zero LIB patches) left it at 0.929. Four anti-collapse architectural fixes failed pre-specified targets. The probes are released alongside the artifact; the full sanity matrix (Appendix I.3, Table 15), applicability matrix per method class (Appendix I.2), and four anti-collapse attempts (Appendix I.3, Table 17) are in the appendix.

6 Discussion

Why the cliff exists. On motion verbs, accuracy is correlated with the held-out verb’s cosine to training (transition zone 0.92–0.94 on B/32, $r=0.627$ on $n=12$). On size and speed, held-out tokens within a class cluster at near-identical cosines but accuracy is determined by class membership: the head defaults to selecting the larger block on size and the slower video on speed under held-out text, with *briskly* at 0.048 ($n=8$ single-token mean) showing the default is strong enough to flip the prediction systematically against the instruction.

This is not a label-imbalance effect (labels are exactly balanced 264/264 on each axis; full audit including an uninformative-text probe is in Appendix J). The head’s class-discriminating direction is fixed by training, but which held-out tokens project onto which side of that direction is mediated by upstream text-encoder geometry: held-out SMALL and FAST tokens project to CLIP-B/32 regions the trained head reads as the opposite class. ViT-L/14 closes both mechanisms by inducing a different text-embedding geometry that places the held-out tokens on the correct side of the learned class direction (motion *transit*: 0.500 $\rightarrow$ 0.917

at L/14 cosine 0.876). Swapping to OpenCLIP-LAION-2B or SigLIP-WebLI (§4.4) only partially closes speed FAST and re-opens cliffs on different tokens, suggesting the head’s class-discriminating direction interacts with each encoder’s specific embedding geometry rather than reflecting a uniform contrastive failure mode.

Implications. Pipelines grounding frozen CLIP-text embeddings through small learned heads—reward models, retrieval, zero-shot classifiers—may be vulnerable to the same text-side transition when lexical surfaces are not varied. Three practical implications follow. **Evaluation:** lexical-held-out splits with cosine-stratified analysis can complement video-held-out splits for diagnosing text-side failure modes that example-level evaluation does not expose. **Detection:** the sanity battery (§5) can help audit multi-pathway architectures. **Mitigation:** our experiments point to three diagnostic interventions—scaling the text encoder (B/32 SMALL 0.485 $\rightarrow$ L/14 0.802), swapping contrastive family (B-B SMALL 0.444 $\rightarrow$ OC-OC 0.929), and test-time text substitution (*transit* 0.500 $\rightarrow$ 0.893)—each diagnostic rather than universally deployable. Characterising the regime where approximate synonym maps suffice, together with prescriptive interventions (Basu et al., 2024), is left to future work.

7 Conclusion

We identify a lexical cliff in compositional binding over frozen CLIP-text embeddings, with two empirically distinct mechanisms, localised primarily to the text encoder via substitution and a 2×2 factorial, and replicated across OpenCLIP-LAION-2B and SigLIP-WebLI with encoder-family-dependent token identity. We recommend lexical-held-out splits with cosine stratification as a complement to standard held-out-example evaluation of small learned heads over frozen CLIP text. CF-PrefBench v4, LIB v0 checkpoints, and probe scripts are released as an [anonymous artifact](#).

Limitations

Per-token analysis is conditional on cue replacement. Per-token cliff analysis (§4.1–§4.5) is scoped to axes whose held-out tokens replace the binding cue (motion, size, speed on v4; action on v3, Appendix G). On axes where held-out tokens only replace wrapper text (color, spatial), no cliff

appears under the standard held-out-lexical split—consistent with the cue-vs-wrapper account in §4. The full 7-axis prediction-vs-outcome matrix is in Appendix F (Table 11). Extending the v4 cosine-stratified probe construction to new axes whose cue tokens admit semantically-graded variants is left to future work.

Backbone scale and encoder-family evidence.

The cliff phenomenon is established on ViT-B/32 and partially closed on ViT-L/14; the cross-family validation (§4.4) extends it across three contrastive systems (OpenAI CLIP, OpenCLIP-LAION-2B, SigLIP-WebLI). We do not claim a smooth monotonic capacity law across CLIP variants. The SigLIP cell additionally varies the contrastive objective (sigmoid vs softmax), patch grid, training data, and preprocessing simultaneously; we treat it as a holistic family swap rather than isolating any single design choice. Whether non-contrastive text encoders (e.g., T5-derived, LLM-derived) or other contrastive variants (EVA-CLIP, MetaCLIP) exhibit similar token-identity-swap patterns is left to future work.

Architecture-independence evidence is effect-size, not fully significance-based. Three architecturally viable trained heads reproduce the cliff at ± 0.05 effect-size consistency, but only the bilinear head reaches $p < 0.05$ under $n=3$ seeds; MLP ($p=0.058$) and cross-attention ($p=0.079$) are marginal (Appendix D). We treat effect-size consistency as the load-bearing claim and flag this as a limitation rather than overclaim significance.

Battery is post-hoc, not predictive. The sanity battery was designed after we caught the Phase 3 false positive on our own architecture, not before. Its current design reflects the specific failure modes we encountered; other modes (train-test split contamination, paraphrase-ID memorisation) would require complementary probes.

Rendered preference simulator and embodied transfer. CF-PrefBench v4 uses a deterministic rendered simulator at 192×144 . The mechanism we identify is text-side, localised via a substitution probe that holds visual input fixed and corroborated by a 2×2 encoder factorial, so the mechanism is independent of visual fidelity. The generalization probe (§4.7, Appendix M) shows the cliff and its text-side mechanism partially re-emerge on embodied rendered video (ManiSkill PushCube,

B/32) but do not transfer to visually distant rendering (CLEVRER); ViT-L/14 does not transfer to ManiSkill, so its cliff is untestable there. The cliff is therefore bounded to where the visual front-end transfers; characterising this boundary across broader embodied and real-world video distributions is the explicit next step.

Simulator-derived labels rather than human preference labels.

Labels are deterministically derived from simulator state, by design for a controlled diagnostic. We do not report inter-annotator agreement because there are no human annotations. Whether human annotators would produce the same labels—and whether the cliff transfers to human-labeled compositional preference data—is an open question.

Ethics Statement

CF-PrefBench v4 trajectories are produced by a deterministic simulator; no human subjects or personal data are involved. Labels are derived from simulator state rather than human annotation, so the benchmark does not aggregate or amplify annotator demographics. The zero-shot judges we compare against (GPT-4o, Qwen2.5-VL-7B) are used solely as baselines, not as training signal. The most plausible dual-use risk is that practitioners run the sanity probes privately without releasing them and continue to publish positive results the battery would have caught. We mitigate this by recommending probe matrices be reported alongside accuracy and by releasing the full probe-runner scripts.

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A Substitution Recovery on Motion Verbs

To test whether the motion-verb cliff is caused by the held-out instruction text or by interaction with the video features, we re-run inference with the held-out instruction’s CLIP text feature replaced by a training-verb instruction with identical direction tokens. The video and label are unchanged.

Verb	Normal	Substituted	Δ
shift	0.917	0.929	+0.012
convey	0.929	0.929	± 0.000
scoot	0.679	0.917	+0.238
transit	0.500	0.893	+0.393

Table 6: Substitution recovery on ViT-B/32 motion verbs (3-seed mean per verb over 84 rows each from the 4-verb cliff probe including *scoot*; arithmetic mean of the four “Normal” values is 0.756, equivalent to the row-level mean since rows are balanced). Recovery scales inversely with cosine, confirming the failure is text-side.

B Expanded Motion-Verb Pool ($n=12$)

To quantify the cosine-correlation relationship in §4.1 on a pool larger than the 4-verb cliff probe, we generated 12 candidate motion verbs and applied strict filters: BPE length ≤ 2 (to control for tokenisation effects), correct part-of-speech in context, SUBTLEX-US Zipf frequency in $[2.0, 6.5]$ (to control for word-frequency confounds), and no overlap with the v4 training or held-out pool. Surviving verbs are stratified to span the CLIP-B/32 cosine range $[0.85, 0.99]$. Spearman $\rho=0.683$ with 95% CI $[0.054, 0.939]$ on the same pool, consistent with the Pearson result reported in the main text. The partial-correlation controls (Zipf, polysemy

via WordNet sense count, BPE length) are applied to the same 12 points.

C Per-Token Cliff Results and Cliff Figures

This appendix collects per-token results across all twelve held-out tokens probed in the main paper. Table 7 reports the $n=8$ per-token means used for the main claims. The Δ column shows the asymmetric closure pattern (large positive Δ on SMALL/FAST/below-cliff cells, near-zero or negative Δ on BIG/SLOW/above-cliff cells). Within-class cosine-independence is visible directly in Table 7: SMALL tokens at $\cos(B/32) \in [0.92, 0.96]$ span accuracy 0.146–0.708, ruling out cosine as the within-class predictor. Figure 5 overlays the ViT-B/32 and ViT-L/14 curves on the 4-verb cliff probe.

Axis	Token	$\cos(B/32)$	B/32	L/14
Motion	shift (above)	0.968	0.917	0.905
	convey (above)	0.938	0.929	0.929
	scoot (below)	0.927	0.679	0.905
	transit (below)	0.915	0.500	0.917
Size	colossal (BIG)	0.898	0.921	0.810
	gigantic (BIG)	0.918	0.929	0.810
	miniature (SMALL)	0.918	0.397	0.817
	petite (SMALL)	0.917	0.492	0.770
	enormous (<i>new</i> , BIG)	0.958	0.923	0.870
	vast (<i>new</i> , BIG)	0.960	0.923	0.870
	tiny (<i>new</i> , SMALL)	0.959	0.708	0.910
	minute (<i>new</i> , SMALL)	0.940	0.146	0.128
Speed	briskly (FAST)	0.937	0.048	0.317
	speedily (FAST)	0.942	0.183	0.460
	sluggishly (SLOW)	0.932	0.960	0.992
	gradually (SLOW)	0.938	0.698	1.000
	swiftly (<i>new</i> , FAST)	0.965	0.247	0.715
	hastily (<i>new</i> , FAST)	0.966	0.202	0.988
	leisurely (<i>new</i> , SLOW)	0.963	0.571	0.682
	languidly (<i>new</i> , SLOW)	0.960	0.946	0.785

Table 7: Per-token cliff results on the expanded $n=16$ -token size/speed pool plus 4 motion tokens, $n=8$ training seeds per backbone. Tokens marked (*new*) added in the expanded-pool revision. Within-class heterogeneity is visible at near-identical cosines: SMALL ranges from *minute* 0.146 to *tiny* 0.708 on B/32 at $\cos \in [0.94, 0.96]$; SLOW exposes *leisurely* 0.571 as a partial cliff (others ≥ 0.68). L/14 closure is *per token* not uniform: *hastily* jumps to 0.988 while *minute* stays at 0.128.

D Robustness Across Alternative Heads (full results)

This appendix gives the full results behind §4.6. We re-train three alternative heads over the same

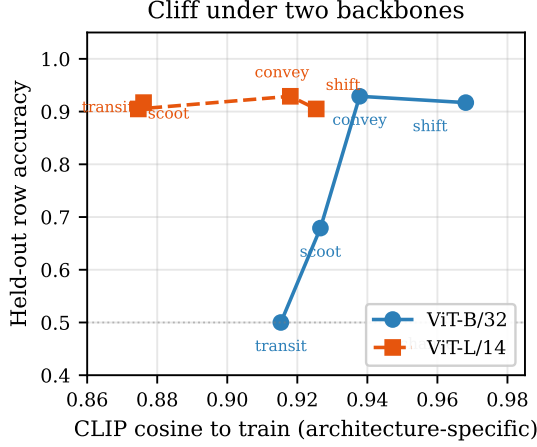


Figure 5: Motion-verb cliff under two backbones. The ViT-L/14 curve (orange, dashed) is flat across all four verbs while ViT-B/32 (blue) exhibits the cliff.

frozen B/32 features (3 seeds each, recipe matched to LIB v0): (i) a bilinear matching head $t^T W v$, (ii) a 2-layer MLP over $[t \parallel \bar{v}]$ (mean-pooled patches), and (iii) a cross-attention head without attribute queries. Two omitted baselines (linear probe, zero-shot CLIPScore) degenerate to chance and are not reported.

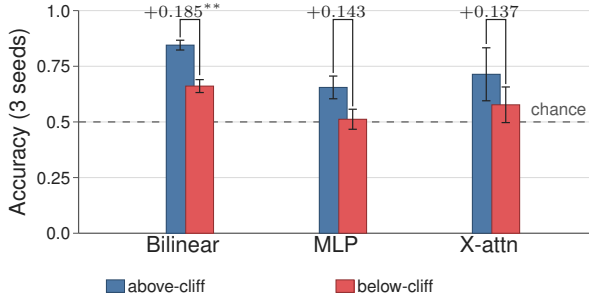


Figure 6: Cliff reproduces on 3/3 architecturally viable learned heads over frozen CLIP-B/32. Bars: 3-seed mean accuracy on above-cliff (blue: {shift, convey}) vs. below-cliff (orange: {scoot, transit}); whiskers ± 1 std. Bilinear gap is significant ($p=0.004$, **); MLP and X-attn are marginal ($p=0.058$, 0.079) under $n=3$. Two omitted baselines (linear probe, zero-shot CLIPScore) degenerate to chance. Effect-size consistency (± 0.05) is the load-bearing claim.

E CF-PrefBench v4 Split Sizes and Paraphrase Pools

This appendix collects the data-side details supporting §3. Table 9 lists per-axis split sizes across the seven binding axes plus the diagnostic impossible_premise split. Table 10 lists the disjoint training and held-out paraphrase pools per

Head	above	below	gap	p
Bilinear	$0.845 \pm .022$	$0.661 \pm .029$	$+0.185$	0.004
2-layer MLP	$0.655 \pm .051$	$0.512 \pm .045$	$+0.143$	0.058
X-attn (no Q)	$0.714 \pm .119$	$0.577 \pm .080$	$+0.137$	0.079

Table 8: Numerical values for Figure 6: above-vs-below degradation across three architecturally viable learned heads over frozen CLIP-B/32 features, 3 seeds each. “X-attn (no Q)” = cross-attention without attribute queries. p from one-sided paired t -test.

axis; the held-out pools are the lexical surface on which the cliff in §4 is measured.

Split	Per-axis size
train	528
val	84
test_seen	72
test_heldout_lexical	84
test_heldout_camera	72
test_heldout_color	84
test_heldout_spatial	84
test_hard_negatives	72

Table 9: Per-split sizes (identical across all 7 binding axes). Plus impossible_premise: 27 rows per non-train split, 0 train. Grand total = $7 \times 1080 + 7 \times 27 = 7,749$.

F Cliff Incidence Across All Seven Axes

CF-PrefBench evolved across two iterations. CF-PrefBench v3 covers four binding axes (color, object, action, spatial) released with disjoint held-out paraphrase pools; CF-PrefBench v4 adds three axes (motion_sequence, size, speed) with cosine-stratified held-out tokens spanning $[0.87, 0.97]$. Table 11 reports row accuracy on test_heldout_lexical alongside the matched test_seen baseline, and classifies each axis by whether its held-out tokens replace the binding cue or only the wrapper text. The cliff signature appears on all four cue-replacement axes and is absent on both wrapper-only axes. Object is intermediate: paraphrase substitution swaps the verb but retains a partially-redundant object noun, giving an attenuated $+0.12$ drop with $PFA > 0$.

G Replication on the v3 Action Axis

The v3 action axis independently exhibits the same cliff mechanism documented for v4 size SMALL and speed FAST. Training-pool verbs are {close, move, open, pull, push}; held-out verbs are {extract, operate, retract, slide}. Held-out instruc-

Axis	Training	Held-out
color	activate, touch	press, engage, tap, trigger
object	bring, move, place	carry, deliver, transport
action	close, move, open, pull, push	extract, operate, retract, slide
spatial	move, place, put	drift, shift, transfer
size	grasp, lift, pick	fetch, retrieve, secure
motion_seq.	drag, move, push	convey, shift, transit, scoot
speed	carry, move, transport	advance, shift, translate

Table 10: Train and held-out paraphrase pools per binding axis. Pools are disjoint by construction. The action axis uses 5 training and 4 held-out verbs (reflecting the v3 release schedule); all other axes use 3 training verbs. The per-token cliff probes for size and speed in §4.5 use eight additional adjective/adverb tokens (BIG/SMALL, FAST/SLOW) outside these verb-paraphrase pools.

tions preserve the noun (*drawer*) and swap only the verb-phrase, which is the binding cue. Train, test_seen, and test_heldout_lexical are disjoint at the video, pair_id, and flip_group levels; labels are balanced 50/50; camera distribution is mixed front+side in both train and test.

H LIB v0 Architecture Details

Let d denote the CLIP text-embedding dimension, $K=8$ the number of sampled frames per video, and P the per-frame patch count ($P=49$ for B/32, $P=256$ for L/14). A linear projection maps the CLIP text embedding to $n=4$ attribute-conditioned query vectors of dimension $d_a=128$ each. The queries attend over the flattened $K \cdot P$ patch tokens through a 4-head multi-head attention layer, followed by LayerNorm and a residual connection. A second linear projection produces n “expected” per-attribute embeddings of the same dimension; cosine similarity between attended and expected embeddings gives a per-attribute binding score in $[-1, 1]$. The 4-d binding vector for each video, together with a 16-d text projection, is concatenated across the two videos and passed through a 64-hidden-unit MLP to yield a scalar preference score. A reconstruction auxiliary—a per-axis classifier head that predicts the instruction’s attribute from the attended embeddings—prevents the attribute queries from collapsing.

The frozen CLIP visual encoder is ~ 87 M params for ViT-B/32 and ~ 305 M for ViT-L/14; trainable parameters total ~ 0.6 M (B/32) or ~ 0.9 M

Ver.	Axis (class)	T.	seen	h-o lex.	Δ	PFA
v3	color	W	0.664	0.824	-0.16	0.50
v3	spatial	W	0.803	0.819	-0.02	0.26
v3	object	W*	0.870	0.753	+0.12	0.60
v3	action	C	0.803	0.413	+0.39	0.00
v4	motion above	C	—	0.902	—	—
v4	motion below	C	—	0.621	—	0.00[†]
v4	size BIG	C	—	0.926	—	—
v4	size SMALL	C	—	0.543	—	—
v4	speed SLOW	C	—	0.826	—	—
v4	speed FAST	C	—	0.092	—	—

Table 11: LIB-B/32 held-out-lexical accuracy across all 7 binding axes, 3-seed mean. T . (Type): **C**=cue-replacement (held-out token replaces the binding cue); **W**=wrapper-only (cue is preserved); **W***=wrapper with redundant object noun. Bold marks cliff cells. [†]Strict PFA on *transit*. **Protocol note:** v4 h-o lex. values here use the original $n=3 \times 2$ -token protocol from the cliff-incidence audit. The canonical $n=8 \times 4$ -token estimates in Table 5 are tighter and slightly different (SMALL 0.485, FAST 0.159); all per-cell claims in §4.5 use the canonical protocol.

Seed	seen acc	h-o acc	Δ acc	h-o PFA
42	0.750	0.381	+0.369	0.000
1337	0.838	0.370	+0.468	0.000
2024	0.819	0.488	+0.331	0.000
mean	0.803	0.413	+0.390	0.000

Table 12: v3 action axis: test_seen vs test_heldout_lexical row accuracy, 3 seeds. PFA collapses to exactly 0 on held-out lexical in every seed.

(L/14). We use $K=8$ frames sampled uniformly across the 24-frame sequence.

I Sanity Battery Details

I.1 Probe Definitions

SC-1 Force gate=1 at inference: the model output is the LIB-only branch; compare to LIB-only baseline. **SC-2** Force gate=0: centroid-only branch; compare to centroid-only baseline. **SC-3** Random gate per call: a random uniform $\sim U[0, 1]$ gate replaces the learned gate. **SC-4** Zero engineered centroid feature: the centroid pair feature is replaced with the zero vector. **SC-5** Zero learned LIB feature: the binding-score vector is replaced with the zero vector. **SC-6** Shuffle CLIP text features within batch: the i -th example’s text feature is replaced by example $\pi(i)$ ’s text feature where π is a random batch permutation.

Held-out verb-phrase	Direction	3-seed acc
retract the drawer	close	0.857
slide the drawer shut	close	0.810
operate the drawer tight	close	0.143
extract the drawer	open	0.310*
slide the drawer ajar	open	0.119
operate the drawer loose	open	0.143

Table 13: v3 action axis: per-held-out-instruction row accuracy (3-seed mean over 14 rows per verb). Close-direction class mean 0.603; open-direction class mean 0.190. The asymmetric default-class bias is structurally identical to v4 size BIG/SMALL (0.924/0.485) and v4 speed SLOW/FAST (0.792/0.159). **extract* shows high seed-variance ($\{0.143, 0.071, 0.714\}$).

I.2 Probe-Applicability Matrix per Method Class

Table 14 summarises which probes apply to each method class in our study.

Method	SC-1	SC-2	SC-3	SC-4	SC-5	SC-6
Engineered centroid	N/A	N/A	N/A	TF	N/A	—
LIB v0 (pure CLIP)	N/A	N/A	N/A	N/A	TF	RUN
Phase 3 hybrid	RUN	RUN	RUN	RUN	RUN	RUN
Zero-shot VLM	N/A	N/A	N/A	N/A	N/A	RUN

Table 14: Probe applicability per method class. TF = trivial-fail (the probe is destructive by construction on this method, so running it gives no diagnostic information). Engineered-centroid SC-6 is omitted because the centroid path parses raw instruction text rather than consuming the CLIP text embedding that SC-6 shuffles.

I.3 Sanity Battery Raw Data per Case Study

Case study 1 — Phase 3 hybrid. Evaluation set: v3 `test_heldout_color` color-axis subset, $n=14$ flip groups, seed 42. The full sanity matrix is shown in Table 15. NORMAL PFA 0.929 = 13/14. SC-1 PFA 0.143 = 2/14. SC-4 PFA 0.167 = 1/6.

Case study 2 — pure-LIB B/32 and L/14. Table 16 reports the relevant probes (SC-6 zero, SC-6 shuffle, substitution) for the pure-LIB architecture on which the cliff was diagnosed in §4.2. SC-6 ZERO collapses both backbones to ≤ 0.500 , confirming the text path is load-bearing. The substitution probe lifts B/32 from 0.750 to 0.893 but cannot move L/14 (already at 0.917), the asymmetry that motivates the text-embedding interpretation in §5.

Case study 3 — Phase 4 anti-collapse. Table 17 reports the per-variant sanity matrix for the four

Probe	PFA	n	Verdict
NORMAL	0.929	14	matches paper claim
SC-1 gate=1	0.143	14	FAIL (well below LIB-only)
SC-2 gate=0	0.929	14	PASS
SC-3 random	0.923	14	gate functionally irrelevant
SC-4 zero cent.	0.167	6	centroid carries 100%
SC-5 zero LIB	0.929	14	LIB unused
SC-6 shuffle txt	0.929	14	text path unused [†]

Table 15: Phase 3 hybrid sanity audit, v3 `test_heldout_color`, $n=14$ flip groups, seed 42. [†]The engineered centroid parses target color separately, so SC-6 understates instruction usage; SC-4/SC-5 is the unambiguous verdict.

Probe	B/32	L/14
NORMAL row acc.	0.750 [‡]	0.917
SC-6 ZERO text	0.500	0.500
SC-6 SHUFFLE text	0.452	—
SUBSTITUTION (train verb)	0.893	0.917

Table 16: Pure-LIB sanity probes on the motion_sequence axis. [‡]Seed-1 retrain on the 3-verb `test_heldout_lexical` split (excludes *scoot*); not directly comparable to the 3-seed 4-verb mean of 0.756 in Table 6.

anti-collapse architectures. Pre-specified targets were NORMAL ≥ 0.75 , SC-5 drop ≥ 0.15 , SC-6 drop ≥ 0.10 ; none of the five variants clears all three.

Variant	NORMAL	SC-5 Δ	SC-6 Δ
B1 bottleneck	0.286	−0.571	+0.071
B2 frozen residual	0.357	n/a	+0.357
B3 MI penalty	0.857	−0.071	+0.087
B4 adv. $\lambda=0.3$	0.786	+0.071	+0.286
B4 adv. $\lambda=0.5$	0.857	0.000	0.000

Table 17: Phase 4 anti-collapse sanity matrix. No variant passes all three pre-specified targets. SC-5 Δ is NORMAL−SC-5 (positive = probe-passing); SC-6 Δ likewise.

J Class-Prior Audit

Training labels on size and speed are exactly balanced (264/264 on each axis), so the default behaviour described in §6 cannot be a label-frequency effect.

We distinguish two probe conditions: (i) **Held-out-class tokens**: under held-out SMALL or FAST tokens (e.g. *briskly*), the head defaults to selecting

the BIG/SLOW video—the dominant §4.5 finding. (ii) **Uninformative text**: replacing the held-out instruction with the embedding of a generic “choose the correct video” string while keeping videos and labels fixed, the trained B/32 model picks BIG on 88% of size rows and the higher-speed video on 100% of speed rows.

The two conditions disagree on speed: held-out FAST tokens elicit SLOW selection, while uninformative text elicits FAST selection. This is consistent with the held-out FAST tokens projecting to text-space regions the head reads as the SLOW class, rather than the head having a unidirectional speed prior. **Caveat**: the uninformative-text result is from a single-seed probe; the held-out-class result (§4.5) is the load-bearing claim, with $n=8$ per cell.

K SigLIP Cross-Family Per-Token Results

Table 18 reports per-token accuracy under SG-SG alongside the matched-protocol B-B ($n=3$, seeds 1/2/3) and OC-OC ($n=3$, seeds 42/123/2024) references. All three cells use architecturally identical LIB v0 heads; SG-SG additionally swaps the pre-projection from 512 $\rightarrow$ 512 to 768 $\rightarrow$ 512 (matching the B-L cell of §4.3). Training recipe is identical across all three cells (60 epochs, AdamW, $\text{lr}_{\text{LIB}}=10^{-4}$, $\text{lr}_{\text{head}}=5 \cdot 10^{-4}$, $\text{lr}_{\text{pre-proj}}=10^{-4}$, weight decay 10^{-4} , dropout 0.3, batch 32, BCE + recon + counterfactual + paraphrase stability losses with weights 1.0/0.1/0.05/0.02).

L Zero-Shot VLM Decodability of Cliff Tokens

To check whether the cliff tokens are decodable under stronger backbones, we evaluate zero-shot GPT-4o-2024-11-20 and Qwen2.5-VL-7B-Instruct on the same 448-row cliff test set, with temperature 0 and a 2×4 image-grid composite of the two video trajectories.

Tie-rate caveat for Qwen2.5-VL-7B. Qwen2.5-VL-7B emits “Tie” on 29–30% of motion/size rows but $< 2\%$ on speed; we score ties as wrong. The speed result is therefore informative: the answer distribution is clean A/B (98%) yet near chance, so the chance-level speed score is a genuine binding failure, not a parsing artifact.

Axis	Token	B-B	OC-OC	SG-SG
Motion	shift	0.917	0.488	0.488
	convey	0.857	0.869	0.833
	scoot	0.679	0.810	0.762
	transit	<i>0.500</i>	0.869	<i>0.798</i>
Size	colossal	0.921	0.968	0.917
	gigantic	0.929	0.976	0.940
	miniature	<i>0.397</i>	0.929	0.921
	petite	<i>0.492</i>	0.929	0.714
Speed	briskly	<i>0.048</i>	0.460	0.865
	speedily	<i>0.183</i>	0.183	<i>0.373</i>
	sluggishly	0.960	0.603	0.333
	gradually	0.698	0.675	1.000

Table 18: Per-token cliff accuracy under three contrastive systems, all $n=3$ seeds. *Italic* marks tokens with class mean ≤ 0.5 . **Bold** marks where the encoder swap changes the qualitative cliff status relative to B-B. Three failure patterns are visible: (i) shared failure across all three systems (*speedily*), (ii) B-B-specific failure closed by both OC-OC and SG-SG (*miniature*, *petite*, *briskly*), and (iii) family-specific new failures (*shift* under OC-OC and SG-SG; *sluggishly* under SG-SG). B-B numbers shown here at $n=3$ differ from the canonical $n=8$ estimates in Table 5; the $n=3$ values are used here for protocol-matched comparison. **Note on B-B convey**: 0.857 here is the 3-seed mean over the protocol-matched seed subset (1/2/3); Table 1 reports 0.929 under the original P1 protocol with a different seed-fitting outcome on seed 2. The difference does not affect any cross-family conclusion.

M Out-of-Domain Rendered Validation

CLEVRER zero-shot evaluation. We constructed 100 disjoint fast/slow pairs from CLEVRER’s (Yi et al., 2020) validation split using per-scene mean velocity (top vs. bottom 30th percentile by characteristic motion). Held-out adverbs {briskly, speedily} (FAST) and {sluggishly, gradually} (SLOW) were tested under the published held-out paraphrase template for the speed axis, with training-pool controls {quickly, slowly} under matched template. All cells, including in-distribution training controls, sit at chance ($\approx 0.49 \pm 0.05$ across B/32 and L/14, 3 seeds): FAST B/32 0.491, SLOW B/32 0.486; FAST L/14 0.523, SLOW L/14 0.437. A substitution probe yields $\Delta \approx 0.0$ pooled across FAST cliff tokens (briskly -0.023 , speedily $+0.002$), confirming the CLEVRER failure is a visual-domain gap, not text-side-recoverable.

ManiSkill pair construction. We use ManiSkill 3.0.1 (Tao et al., 2024) PushCube-v1 (Franka Panda + cube on tabletop), rendered headless at

Axis	Class	Qwen-7B	Tie	GPT-4o
Motion	above-cliff	0.357	0.286	0.538
	below-cliff	0.446	0.161	0.539
Size	BIG	0.500	0.298	0.890
	SMALL	0.440	0.286	0.810
Speed	FAST	0.452	0.012	0.680
	SLOW	0.452	0.012	0.550

Table 19: Per-class zero-shot VLM accuracy on the cliff test set (two tokens per class). For Qwen, “Tie” is the fraction of trials where the model refused to commit to A or B (scored as wrong; speed has $< 2\%$ ties). GPT-4o produced no explicit ties; accuracies computed on parseable A/B answers (9/112 motion and $\leq 6/168$ size trials unparseable; speed had none, so its chance-level speed score is not a parsing artifact). GPT-4o has a complementary failure mode: at 192×144 resolution it cannot reliably read direction tokens, scoring 0.48–0.60 on all four motion verbs including those where LIB scores 0.92.

on L/14: FAST control 0.567 [0.507, 0.630], SLOW control 0.553 [0.490, 0.617], FAST held-out 0.538 [0.470, 0.605], SLOW held-out 0.538 [0.482, 0.597]. The cliff is therefore untestable under L/14 on ManiSkill. We do not report L/14 substitution probes because the control is at chance, making any recovery uninterpretable.

Reproducibility. 100 pair seeds, 3 LIB seeds per backbone, action-scale $\{1.0, 0.3\}$, all parameters fixed a priori before LIB inference. No retraining; published LIB checkpoints used zero-shot throughout. All scripts, 5 QC mp4 visualisations demonstrating the $\times 1.0 / \times 0.3$ speed gap, and per-trial predictions are released alongside the artifact.

224×224 via the bundled Vulkan ICD. For each pair, we sample one 24-step action sequence using the environment’s random policy (seed= $17i + 31$ for pair i , deterministic), then replay the *same* action sequence from the *same* initial state at two action amplitudes: $\times 1.0$ (fast) and $\times 0.3$ (slow). This yields paired videos whose only difference is motion magnitude. Quality control on 100 pairs: mean end-effector displacement per frame 0.0184 (fast) vs. 0.0055 (slow), a $3.3\times$ gap on 100/100 pairs. The cube is rarely directly contacted by the random policy, so the speed signal is arm/scene pace rather than cube motion; this is documented as a scope limitation.

Cross-protocol reference. The simulator B/32 baseline at matched $n=3$ protocol (Table 4) is FAST 0.115 / SLOW 0.829; canonical $n=8$ estimates (Table 5) are FAST 0.159 / SLOW 0.792.

Per-token results on ManiSkill (B/32, 3 seeds). Training tokens: “move the block quickly” 0.640 [0.570, 0.707]; “move the block slowly” 0.680 [0.610, 0.743]. Held-out FAST: briskly 0.327, speedily 0.333 (class mean 0.330 [0.267, 0.387]). Held-out SLOW: sluggishly 0.640, gradually 0.590 (class mean 0.615 [0.562, 0.668]). Substitution probe (FAST adverb embedding $\rightarrow$ “move the block quickly”; video and label fixed): briskly $0.327 \rightarrow 0.640$ ($\Delta=+0.313$), speedily $0.333 \rightarrow 0.640$ ($\Delta=+0.307$), pooled $\Delta=+0.310$.

ManiSkill L/14 null result. Both training-token control and held-out tokens sit near chance